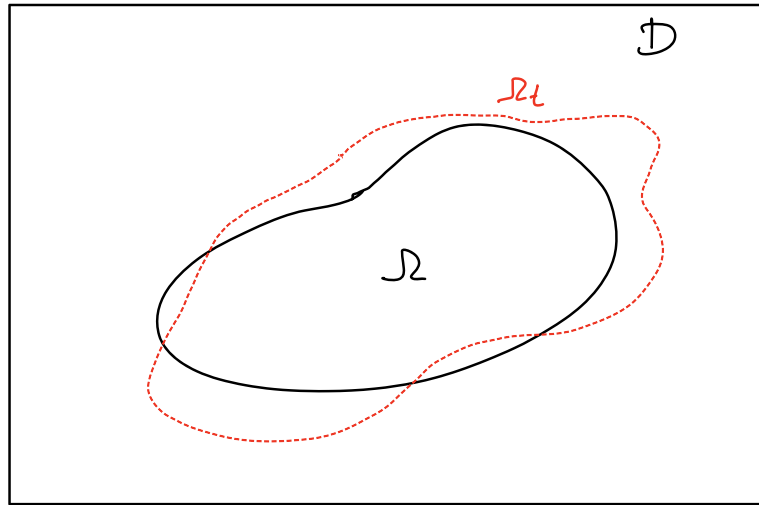


## On shape Hessians



$\Omega \subset D$  ("holdall") subject to shape optimization

Most common strategy for shape changes: Transformations

$$\Omega_t = T_t(\Omega) \quad t \in \mathbb{R} \quad \Omega_0 = T_0(\Omega) = \Omega$$

Examples: (i) free-form deformations

Grid of control points in  $D$  that deforms  $D$  through interpolating polynomials

Design variables:

location of control points

(ii) perturbation of identity

For  $x \in \Omega_t$ ,

$$T_{tr}(x) = x + t V(x)$$

$V: D \rightarrow \mathbb{R}^3$  ("velocity"),  $t \in \mathbb{R}$

More flexible and common than (i)

(iii) goes well with mesh deformations

Important:  $V$  a function of  $x$ , not constant!

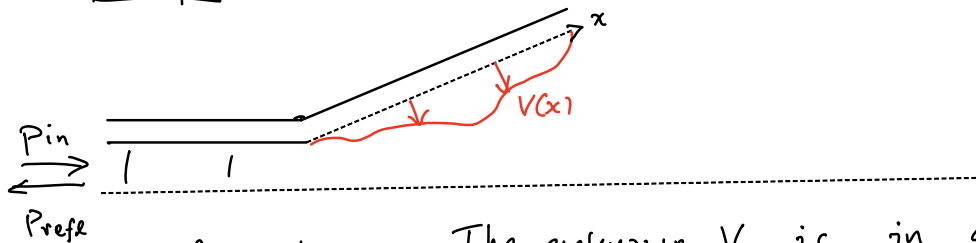
(Otherwise rigid body motion; no shape changes!)

Now consider (ii), deformations through perturbations of identity

Two approaches (a) optimize over displacements from a fixed reference frame

Objective: find the optimal displacement ("velocity") field  $V$ ;  $\Omega = \Omega_{\text{ref}}$  fixed

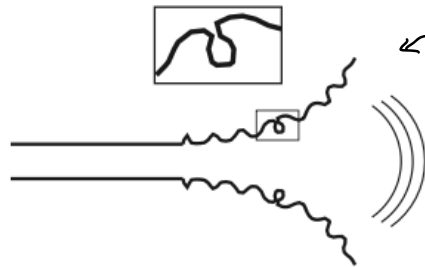
Example: our first horn articles ( $\Omega_{\text{ref}}$ : a conical horn)



Advantage: The unknown  $V$  is in a linear space. Any classical optimization algorithm suitable, e.g.

Newton, BFGS quasi-Newton

Limitation: Constraints on feasible shapes



Turns out to be better but not reachable w/ approach (a) !

Approach (b): At each optimization iteration, update from previous shape

$$T_{tV}(x) = x + tV(x)$$

$$\Omega_{n+1} = T_{t_n V_n}(\Omega_n)$$

Ex:  $T_{tV}(x) = x + tV(x) = \hat{x}$

$$\begin{aligned} T_{sW}(T_{tV}(x)) &= T_{sW}(\hat{x}) = \hat{x} + sW(\hat{x}) \\ &= x + tV(x) + sW(x + tV(x)) \end{aligned}$$

$$\neq x + t V(x) + s W(x)$$

(unless  $W$  is constant)

- Has profound consequences for Hessian computations!
- Consequences for optimization algorithms:
  - (a) Each step a "new" optimization problem!
  - (b) Not a problem for first-order methods  
Shape gradient is linear in velocity  $V$
  - (c) Problematic for BFGS quasi-Newton!
  - (d) Newton's method works! — with "the right" Hessian

Recall: Newton's method for minimization

Works by minimizing, at each iteration, a model quadratic objective. Ex:  $\min_{x \in \mathbb{R}^n} f(x)$

$$f(x_n + s) \approx F_n(s) = f(x_n) + \underbrace{s \cdot g(x_n)}_{\text{linear in } s} + \frac{1}{2} \underbrace{s \cdot H(x_n) s}_{\text{bilinear in } s}$$

$$\text{Minimizing } F_n \Rightarrow H(x_n) s = -g(x_n)$$

$$e_i \cdot H(x_n) e_j = \frac{\partial^2}{\partial s \partial t} f(x_n + t e_i + s e_j) \Big|_{t=s=0}$$

For shape optimization  $\min_{\Omega} J(\Omega)$

At current iteration  $\Omega$ . Want to find velocity

field  $V$  minimizing a model quadratic functional in  $V$ :

$$J(T_V(\Omega)) \approx j_{\Omega}(V) = J(\Omega) + \underbrace{dJ(\Omega; V)}_{\text{linear in } V} + \frac{1}{2} \underbrace{d^2 J(\Omega; V, V)}_{\text{bilinear in } V}$$

What is  $dJ$ ?

$$dJ(\Omega; V) = \lim_{t \rightarrow 0} \frac{1}{t} [J(T_{tV}(\Omega)) - J(\Omega)] = \frac{d}{dt} J(T_{tV}(\Omega)) \Big|_{t=0}$$

$J$  is shape differentiable if the limit exists for all admissible  $V$  and if mapping  $V \mapsto dJ(\Omega; V)$  is linear and continuous.

What is  $d^2 J$ ?

$$d^2 J(\Omega; V, W) = \frac{\partial^2}{\partial s \partial t} J(T_{tV+sW}(\Omega)) \Big|_{t=s=0} \quad (1)$$

$J$  is twice shape differentiable if the limit exists for all adm.  $V, W$ , and if  $V, W \mapsto dJ(\Omega; V, W)$  is bilinear and continuous.

We are used to that the second derivative is the derivative of the first derivative.

What about  $d^2 J$ ? Compare with

$$\begin{aligned} ddJ(\Omega; V, W) &= \lim_{s \rightarrow 0} \frac{1}{s} [dJ(T_{sW}(\Omega); V) - dJ(\Omega; V)] \\ &= \frac{d}{ds} dJ(T_{sW}(\Omega); V) \Big|_{s=0} \\ &= \frac{d^2}{ds dt} J(T_{sW}(T_{tV}(\Omega))) \Big|_{t=s=0} \quad (2) \end{aligned}$$

Recall from above that

$$\begin{aligned} T_{sW}(T_{tV}(x)) &= x + tV(x) + sW(x + tV(x)) \\ &\neq x + tV(x) + sW(x) \end{aligned}$$

So the  $d^2 J$  suitable for Newton's method is not  $ddJ$ ! However, it holds that

$$d^2 J(\Omega; V, W) = ddJ(\Omega; V, W) - dJ(\Omega; (W \cdot V)V) \quad (3)$$

Example:  $J_1(\Omega) = \int_{\Omega} f dV$   $f, g: D \rightarrow \mathbb{R}$   
smooth, fixed  
 $J_2(\Omega) = \int_{\partial\Omega} g dS$

We know that  $dJ_1(\Omega; V) = \int_{\partial\Omega} f n \cdot V dS \quad (4)$

$$dJ_2(\Omega; V) = \int_{\partial\Omega} \left[ \frac{\partial g}{\partial n} + \kappa g \right] n \cdot V dS \quad (5)$$

where  $\kappa$  is the summed curvature field on  $\partial\Omega$

$[\kappa = \nabla_\Gamma \cdot n]$ . Thus, since  $dJ_1(\Omega; V) = \int_{\partial\Omega} f n \cdot V dS$ ,

$$\begin{aligned} \text{by (5), } d d J_1(\Omega; V, W) &= \int_{\partial\Omega} \left[ \frac{\partial}{\partial n}(f n \cdot V) + \kappa f n \cdot V \right] n \cdot W dS \\ &= \int_{\partial\Omega} \left[ \frac{\partial}{\partial n}(f n \cdot V) n \cdot W + \kappa f n \cdot V n \cdot W \right] dS \quad (5) \end{aligned}$$

Recall (3):

$$\begin{aligned} d^2 J_1(\Omega; V, W) &= d d J_1(\Omega; V, W) - d J_1(\Omega; (W \cdot \tau) V) \\ &= \int_{\partial\Omega} \left[ \frac{\partial}{\partial n}(f n \cdot V) n \cdot W + \kappa f n \cdot V n \cdot W \right] dS - \int_{\partial\Omega} f n \cdot [(W \cdot \tau) V] dS \quad (6) \end{aligned}$$

$$\begin{aligned} \text{Now, } n \cdot [(W \cdot \tau) V] &= n \cdot \left[ W \cdot \left( n \frac{\partial}{\partial n} + \nabla_\Gamma \right) V \right] \\ &= n \cdot W \frac{\partial}{\partial n}(n \cdot V) + 0 \end{aligned}$$

$\uparrow$  since  $n \cdot \nabla_\Gamma = 0$

$$\Rightarrow - \int_{\partial\Omega} f n \cdot [(W \cdot \tau) V] dS = - \int_{\partial\Omega} f n \cdot W \frac{\partial}{\partial n}(n \cdot V) dS \quad (7)$$

Combining (6) & (7) yields

$$d^2 J_1(\Omega; V, W) = \int_{\partial\Omega} \left[ \frac{\partial f}{\partial n} + \kappa f \right] n \cdot V n \cdot W dS$$

Note:  $d^2 J_1$  is symmetric in  $V, W$  but  $d d J_1$  is not!